

1. The expression $n \& (n-1)$ will turn off the rightmost set bit of a number n . The expression $n-1$ will have all the bits flipped after the rightmost set bit of n (including the rightmost set bit). So, $n \& (n-1)$ will result in the last bit flipped of n . Consider,

00010100 & (n = 20)

00010011 (n-1 = 19)

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00010000

00...00000000      &      (n = 0, no rightmost bit)

11...11111111      (n-1 = -1)

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00...00000000

Now, using this idea above, find the total number of bits needed to be flipped to convert a given integer to another.

Example:

Input:

`x = 65 (which is 01000001 in binary)`

`y = 80 (which is 01010000 in binary)`

Output: `The total number of bits to be flipped is 2`